

Initial Sorting into Schools: Distance, Ability, and some Implications for the Model

April 2026

1 Setup

To study how students sort into schools at grade 1, I construct a dataset of student–school pairs. Each student is matched to all 83 schools in the sample. Distance is computed from the student’s **home address** to each school’s location.

Table 1 shows that students attend nearby schools: the average distance to the chosen school is 5.7 km, and 50% of students attend one of their three closest schools.

Table 1: How Close Are Chosen Schools?	
Distance	
Average to chosen school	5.7 km
Average to non-chosen schools	44.2 km
Proximity ranking of chosen school	
Closest school	21.8%
Among 3 closest	49.8%
Among 5 closest	67.4%
Among 10 closest	85.8%

Schools differ along two dimensions relevant for the model¹. *School type* captures tracks offered: ASO-only (37%), AUT/MS or cycle-1-only (24%), multilateral (20%), and TSO/BSO-only (19%). *Cycle length* determines how long students can stay: 41% attend cycle-1-only schools (grades 1–2 only, must switch after), while 54% attend full-cycle (1+2+3) schools.

2 Results

I estimate a conditional logit of school choice with three specifications on the same sample ($N = 78,615$ student–school pairs, 4,430 students²).

The choice set is restricted to *feasible* schools: I remove gender-infeasible schools (boys cannot choose girls-only schools and vice versa) and schools beyond 20 km from the student’s home. Results are robust to 10 km and 30 km cutoffs .

¹Here I am abstracting from school characteristics aside from these two dimensions. That is, I am not considering “vertical” quality differences within ASO or TSO/BSO schools. For future iterations, I am considering adding measures of school quality, proxied by neighborhood demographics and school-level survey data, to separate track preferences from vertical quality indicators - but likely it will only strengthen the argument.

²These students are representative of the in estimation of the advice model (on average have the same observables).

Table 2: Conditional Logit: School Choice at Grade 1

	(1)	(2)	(3)
Distance (km)	−0.362*** (0.006)	−0.374*** (0.006)	−0.379*** (0.006)
School characteristics	No	Yes	Yes
Student × school interactions	No	No	Yes
Log-likelihood	−8,514	−8,033	−7,273
Observations	78,615	78,615	78,615
Students	4,430	4,430	4,430
LR test (2 vs. 3)			$\chi^2(54) = 1,520^{***}$

Notes: Conditional logit. Unit of observation: student–school pair. School characteristics: cycle length (1, 1+2, 1+2+3, 1+2+3+4) and school type (ASO-only, AUT/MS, multilateral, TSO/BSO-only). Student characteristics interacted with school characteristics include: verbal IQ, numeric IQ (both continuous), gender, parental income (5 categories), and parental education (3 levels). Choice set restricted to gender-feasible schools within 20 km. Standard errors in parentheses. *** $p < 0.01$.

Distance dominates. Each additional kilometer reduces the odds of choosing a school by approximately 30%³. The coefficient is stable across all specifications.

But ability sorting exists beyond geography. Adding student × school interactions (column 3) significantly improves fit (LR $\chi^2(54) = 1,520$, $p < 0.001$). To quantify the magnitude of sorting relative to distance, I express interaction coefficients as *distance equivalents*: dividing each by $|\beta_{\text{distance}}| = 0.379$ yields the kilometers of distance that would produce the same effect on school choice. Table 3 reports the key results.

³I try specifications with quadratic distance and log distance, but for interpretability keep the linear one. Actually, quadratic is rejected in favor of linear in this restricted choice setting.

Table 3: Ability Sorting: Distance-Equivalent Effects

Student characteristic	Comparison	Coeff.	Dist. equiv.
<i>Verbal IQ (per 10 points) × School type (ref: ASO-only)</i>			
+10 pts	→ AUT/MS	−0.357***	+0.9 km further
+10 pts	→ Multilateral	−0.383***	+1.0 km further
+10 pts	→ TSO/BSO	−0.696***	+1.8 km further
<i>Numeric IQ (per 10 points) × School type (ref: ASO-only)</i>			
+10 pts	→ AUT/MS	−0.122**	+0.3 km further
+10 pts	→ Multilateral	−0.346***	+0.9 km further
+10 pts	→ TSO/BSO	−0.581***	+1.5 km further
<i>Numeric IQ (per 10 points) × Cycle length (ref: cycle 1 only)</i>			
+10 pts	→ Cycle 1+2+3	+0.269***	−0.7 km closer
+10 pts	→ Cycle 1+2+3+4	+0.345***	−0.9 km closer
<i>Parental education (ref: lowest) × School type (ref: ASO-only)</i>			
Highest level	→ AUT/MS	−0.896***	+2.4 km further
Highest level	→ Multilateral	−0.493***	+1.3 km further
Highest level	→ TSO/BSO	−1.035***	+2.7 km further
<i>Gender (ref: female) × School type (ref: ASO-only)</i>			
Male	→ AUT/MS	+0.354***	−0.9 km closer
Male	→ Multilateral	−0.366***	+1.0 km further
Male	→ TSO/BSO	+1.375***	−3.6 km closer

Notes: Selected interactions from column 3 of Table 2. Distance equivalent = $\beta_{\text{interaction}}/|\beta_{\text{distance}}|$. “Further” means the student treats the school type as if it were that many km more distant than an ASO-only school; “closer” means the opposite. IQ effects are per 10-point increase. For reference, the average distance to the chosen school is 5.7 km. Only significant interactions shown ($p < 0.05$); full results available in the appendix.

The sorting patterns are clear and monotonic. Higher-IQ students avoid non-ASO schools: a student with 30 points higher verbal IQ (roughly 2 standard deviations) treats a TSO/BSO school as if it were 5.5 km further away than an ASO-only school. Given that the average commute is 5.7 km, this is large enough to substantially reduce the probability of attending TSO/BSO for high-ability students. Verbal IQ produces stronger sorting than numeric IQ, and TSO/BSO schools show the strongest ability gradient, followed by multilateral and AUT/MS schools.

Higher numeric IQ also predicts preference for longer-cycle schools: a 10-point increase makes full-cycle (1+2+3) schools 0.7 km closer in equivalent terms, consistent with higher-ability students seeking institutions where they can complete their full secondary education without switching.

Parental education shows a strong and consistent pattern: students with the highest parental education level treat TSO/BSO schools as 2.7 km further and AUT/MS schools as 2.4 km further than ASO-only schools. This operates through school type rather than cycle length, suggesting that parental education captures preferences over academic orientation beyond what ability alone reflects.

Gender sorting is concentrated in TSO/BSO schools, where boys are 3.6 equivalent-km more likely to attend, consistent with anticipation of STEM and technical orientations –which lead to male dominated jobs. Interestingly, boys are *less* likely to attend multilateral schools (1.0 km further), suggesting that gender sorting operates primarily through the vocational/technical dimension.

2.1 Summary

Distance is the primary driver of school choice, but there is economically meaningful ability-based sorting into school types that is not explained by geography alone. Higher-IQ and higher-parental-education students differentially select into ASO-only and longer-cycle schools. The sorting is monotonic across school types (strongest away from TSO/BSO, weakest for AUT/MS) and cumulative across ability dimensions (IQ and parental education reinforce each other). These patterns imply that the initial distribution of student types across schools is not independent of school characteristics.

3 Implications for the Model -some thoughts

3.1 Handling initial selection

The distribution of student types across schools at entry is not random: the results show that $F(\theta|s)$ depends on school type s . This means cross-school comparisons of recommendation behavior are potentially confounded by selection: a school may recommend differently not because it is strategic, but because it serves a different student population. Two approaches could potentially address this:

1. **Conditioning on latent types.** Recover student types from the measurement system (IQ, test scores, teacher assessments, repeated across time) using BLM/Hu-Schennach or Hu-Shum methods. Since IQ is both the main driver of sorting (as shown above) and a key input to the measurement system, conditioning on recovered types should absorb much of the selection. The residual variation in recommendations across schools, conditional on estimated types, should identify the school's strategic behavior. However, this approach assumes the measurement system captures all relevant dimensions of heterogeneity. Unobserved factors such as parental preferences for academic paths/parental effort⁴ — which may drive school choice/recommendations but may not appear in IQ or test scores — would remain uncontrolled.
2. **Joint modeling.** Include the grade-1 school choice as the first period of the structural model. Distance plausibly provides exogenous variation in school choice, conditional on observables⁵. The latent type enters both the school choice utility and the recommendation model, so the joint estimation simultaneously identifies the type distribution and the school's recommendation strategy. Distance serves as an excluded variable: it shifts school choice but does not directly affect recommendations. This is probably more demanding but provides cleaner identification, as it does not require the measurement system to capture all dimensions of selection.

Maybe a natural path forward is to start with approach 1 as a first pass, then check robustness by adding the joint school choice model (approach 2).

3.2 Identifying the school's strategic motive

Finally, note that the data offers a natural source of identification to separate how schools use advice. Nearly 41% of students attend **cycle-1-only schools**, which do not retain students beyond grade 2. These schools have no capacity constraints for later grades and no retention incentive: they cannot benefit from steering students toward particular tracks, because the student leaves regardless. In contrast, schools offering cycles 1+2+3 retain students and face direct incentives tied to pass rates, capacity, and track composition.

Comparing recommendation patterns at cycle-1 schools (we might assume "weaker" or different strategic incentive) versus 1+2+3 schools, conditional on student ability, could directly identify the strategic distortion. Cycle-1 schools may provide a benchmark for purely informational recommendations. The gap between this benchmark and the behavior of 1+2+3 schools would quantify the cost of strategic information provision. This is a tentative idea, to be explored empirically...

3.3 Do parents anticipate tracking policy?

A key question for the structural model is whether students' enrollment decisions respond to schools' tracking policies. If parents choose schools anticipating how the school will certificate and recommend their child at the end of grade 2, then school choice is a forward-looking decision that must be jointly

⁴We may have proxies for this, but only measured at baseline.

⁵While distance is treated as an exogenous constraint, it is clear that residential location can be an endogenous choice made by parents. If high-SES parents move closer to ASO schools, the distance coefficient may partially capture these latent preferences. However, the stability of the coefficient across specifications suggests distance remains a robust primary constraint.

modeled with the recommendation stage. If, instead, parents choose schools based on observable characteristics — type, distance, peer quality — without anticipating tracking policy, enrollment can be treated as a simple myopic reduced-form first stage.

To test this, I add the school’s historical A-certificate rate to the conditional logit. This variable captures the percentage of students receiving an A certificate (unrestricted pass) at the end of grade 2, reported by each school for a cohort preceding our sample⁶. It is reasonable to assume this information is available to parents when making their enrollment decision, either directly or through reputation.

The A-certificate rate is an imperfect measure of tracking leniency: schools with higher-ability students mechanically produce more A certificates regardless of their policy. To address this, I control for peer composition using the leave-one-out mean numeric IQ at each school — computed from our cohort but excluding the student in question. This assumes the distribution of ability across schools is approximately stable across connected cohorts, which is plausible given the geographic nature of school assignment that I document above.

I report results in Table 4. In column (A), I add the A-certificate rate⁷ as a school characteristic to the baseline specification from Table 2. The coefficient is positive but economically negligible: a 10 percentage point increase in the A-certificate rate is equivalent to only 0.1 km of distance — this is trivial relative to the 5.7 km average commute. In column (B), I additionally control for peer numeric IQ. The certificate rate coefficient flips sign but remains negligible in magnitude (0.1 km). In contrast, peer composition has a substantial effect: a 10-point increase in peer numeric IQ is equivalent to 2.2 km of distance.

These results suggest that parents respond to peer composition but not to tracking policy when choosing schools. The A-certificate rate — whether interpreted as leniency, strictness, or quality — has no economically meaningful effect on enrollment in either specification. This supports treating enrollment as myopic with respect to tracking policy, which considerably simplifies the first stage of the structural model.

Table 4: Do Students Respond to Tracking Policy?

	(A) + Cert. rate	(B) + Cert. rate & peer quality
Distance (km)	−0.379*** (0.006)	−0.389*** (0.006)
% A certificates (prior cohort)	+0.005*** (0.001)	−0.005*** (0.001)
<i>Dist. equiv. (per 10 pp)</i>	<i>−0.1 km</i>	<i>+0.1 km</i>
Peer numeric IQ (leave-one-out)		+0.086*** (0.004)
<i>Dist. equiv. (per 10 pts)</i>		<i>−2.2 km closer</i>
School characteristics	Yes	Yes
Student × school interactions	Yes	Yes
Log-likelihood	−7,173	−6,922
Observations	76,467	76,467

Notes: Both columns include the full set of school type, cycle length, and student × school interactions from Table 2. “% A certificates” is the school’s historical share from a prior cohort. “Peer numeric IQ” is the leave-one-out school mean. Distance equivalents: $10 \times \beta / |\beta_{\text{distance}}|$. The certificate rate effect is economically negligible (≈ 0.1 km) in both specifications, while peer composition has a substantial effect (2.2 km per 10 IQ points). Average commute: 5.7 km.

⁶It is actually the average percentage of students that have got an A certificate in the last 3 years in this school and grade.

⁷Note that in the data there seems to be some variation in this measure. Mean % of A certificates is 78% and standard deviation is 14.02,

4 Appendix

Table 5: Full Conditional Logit Results (Column 3)

	Coefficient
Distance	
Distance (km)	-0.379*** (0.006)
Cycle length (ref: Cycle 1 only)	
Cycle 1+2	-1.527* (0.861)
Cycle 1+2+3	-2.051*** (0.419)
Cycle 1+2+3+4	-2.531*** (0.944)
School type (ref: ASO-only)	
AUT/MS	5.080*** (0.564)
Multilateral	7.194*** (0.512)
TSO/BSO	11.389*** (0.550)
Cycle length $\times$ Household income (ref: lowest)	
Cycle 1+2 $\times$ Income 2	-0.196 (0.314)
Cycle 1+2 $\times$ Income 3	-0.023 (0.346)
Cycle 1+2 $\times$ Income 4	-0.169 (0.426)
Cycle 1+2 $\times$ Income 5	-0.053 (0.574)
Cycle 1+2+3 $\times$ Income 2	-0.056 (0.137)
Cycle 1+2+3 $\times$ Income 3	0.025 (0.142)
Cycle 1+2+3 $\times$ Income 4	-0.189 (0.160)
Cycle 1+2+3 $\times$ Income 5	0.103 (0.182)
Cycle 1+2+3+4 $\times$ Income 2	-0.425 (0.291)
Cycle 1+2+3+4 $\times$ Income 3	-0.528* (0.321)
Cycle 1+2+3+4 $\times$ Income 4	-0.590 (0.439)
Cycle 1+2+3+4 $\times$ Income 5	-0.218 (0.607)
Cycle length $\times$ Verbal IQ	
Cycle 1+2 $\times$ Verbal IQ	-0.004 (0.010)
Cycle 1+2+3 $\times$ Verbal IQ	-0.001

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	Coefficient
Cycle 1+2+3+4 $\times$ Verbal IQ	(0.004) 0.010 (0.010)
Cycle length $\times$ Numeric IQ	
Cycle 1+2 $\times$ Numeric IQ	0.008 (0.010)
Cycle 1+2+3 $\times$ Numeric IQ	0.027*** (0.004)
Cycle 1+2+3+4 $\times$ Numeric IQ	0.034*** (0.011)
Cycle length $\times$ Male	
Cycle 1+2 $\times$ Male	0.089 (0.244)
Cycle 1+2+3 $\times$ Male	-0.048 (0.098)
Cycle 1+2+3+4 $\times$ Male	-17.201 (439.999)
Cycle length $\times$ Parental education (ref: no degree)	
Cycle 1+2 $\times$ High school	-0.700** (0.321)
Cycle 1+2 $\times$ Higher educ.	-1.107*** (0.403)
Cycle 1+2+3 $\times$ High school	-0.042 (0.123)
Cycle 1+2+3 $\times$ Higher educ.	-0.313** (0.140)
Cycle 1+2+3+4 $\times$ High school	-0.119 (0.280)
Cycle 1+2+3+4 $\times$ Higher educ.	-0.848** (0.432)
School type $\times$ Household income (ref: lowest)	
AUT/MS $\times$ Income 2	0.487*** (0.171)
AUT/MS $\times$ Income 3	0.329* (0.172)
AUT/MS $\times$ Income 4	-0.053 (0.202)
AUT/MS $\times$ Income 5	0.090 (0.228)
Multilateral $\times$ Income 2	0.550*** (0.147)
Multilateral $\times$ Income 3	0.108 (0.150)
Multilateral $\times$ Income 4	0.201 (0.176)
Multilateral $\times$ Income 5	-0.226 (0.206)
TSO/BSO $\times$ Income 2	0.476*** (0.152)
TSO/BSO $\times$ Income 3	0.182 (0.157)

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	Coefficient
TSO/BSO $\times$ Income 4	0.108 (0.206)
TSO/BSO $\times$ Income 5	-0.400 (0.271)
School type $\times$ Verbal IQ	
AUT/MS $\times$ Verbal IQ	-0.036*** (0.006)
Multilateral $\times$ Verbal IQ	-0.038*** (0.005)
TSO/BSO $\times$ Verbal IQ	-0.070*** (0.005)
School type $\times$ Numeric IQ	
AUT/MS $\times$ Numeric IQ	-0.012** (0.005)
Multilateral $\times$ Numeric IQ	-0.035*** (0.005)
TSO/BSO $\times$ Numeric IQ	-0.058*** (0.005)
School type $\times$ Male	
AUT/MS $\times$ Male	0.354*** (0.121)
Multilateral $\times$ Male	-0.366*** (0.108)
TSO/BSO $\times$ Male	1.375*** (0.130)
School type $\times$ Parental education (ref: no degree)	
AUT/MS $\times$ High school	-0.519*** (0.152)
AUT/MS $\times$ Higher educ.	-0.896*** (0.172)
Multilateral $\times$ High school	-0.290** (0.130)
Multilateral $\times$ Higher educ.	-0.493*** (0.153)
TSO/BSO $\times$ High school	-0.379*** (0.137)
TSO/BSO $\times$ Higher educ.	-1.035*** (0.181)
Model fit	
Log-likelihood	-7,273
Pseudo R^2	0.414
Observations	78,615
Students	4,430

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reference categories: ASO-only (school type), Cycle 1 (cycle length), lowest income (household income), female (gender), no degree (parental educ.).